

SAT Hard Question 5

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Question: Expression

The given expression can be rewritten as $\frac{a}{4}x^2 + \frac{b}{4}x + \frac{c}{4}$ where a , b , and c are constants. What is the value of $a + b + c$? Given the expression $-16(5x - 3)^2 + 4(5x - 2)^2$.

A -112

B -56

C -28

D -7

Answer

Correct Answer: A.

The coefficients a , b , and c are derived from the given quadratic expression by expanding and combining like terms. Calculating $a + b + c$ yields -112 .

Question 20

In the given function, b is a positive constant. For every increase in the value of x by 1, the value of $f(x)$ increases by $c\%$ where $0 < c < 100$. Which expression gives the value of c in terms of b ?

$$f(x) = 66(b)^x$$

- A. $1 + \frac{b}{100}$
- B. $b + 100$
- C. $100(b + 1)$
- D. $100(b - 1)$

Answer to Question 20

Correct Answer: D.

Given the exponential function $f(x)$, the percentage change for an increment of 1 in x corresponds to $100(b - 1)\%$.

Question 11

If $\sin x^\circ = a$, which of the following must be true for all values of x ?

A. $\cos x^\circ = a$

B. $\sin(90^\circ - x^\circ) = a$

C. $\sin(x^2)^\circ = a^2$

D. $\cos(90^\circ - x^\circ) = a$

Answer to Question 11

Correct Answer: D.

This is the complementary angle identity: $\sin(\theta) = \cos(90^\circ - \theta)$.

Question 20

The function g is a quadratic function in the xy -plane. The graph of $y = g(x)$ has a vertex at $(-1, -4)$ and passes through the points $(-2, -43)$ and $(1, -160)$. What is the value of $g(0) - g(2)$?

- A. 312
- B. 355
- C. -117
- D. -203

Answer to Question 20

Correct Answer: A.

Question 22

The mass of object A is 324% of the mass of object B and the mass of object A is 7.2% of the mass of object C. If the mass of object C is $p\%$ of the mass of object B, what is the value of $p\%$?

- A. 4.5%
- B. 4.5
- C. 45
- D. 450

Answer to Question 22

Correct Answer: C.

Question 22

In the given equation, c is a constant. If the equation has exactly one solution, what is the value of c ?

$$\frac{1}{cx} = \frac{x}{76} + \frac{1}{c}$$

Answer to Question 22

Correct Answer: -19.

Question 22

A parallel electric circuit contains four resistors with resistances of p ohms, q ohms, s ohms, and 13 ohms. The given equation relates the resistance R , in ohms, of the circuit, to the resistances of the four individual resistors it contains. Which equation correctly expresses R in terms of p , q , and s ?

$$\frac{1}{R} = \frac{1}{p} + \frac{1}{q} + \frac{1}{s} + \frac{1}{13}$$

- A. $R = \frac{13pqs}{p+q+s+13}$
- B. $R = \frac{13pqs}{13qs+13ps+13pq+pqs}$
- C. $R = \frac{13qs+13ps+13pq+pqs}{13pqs}$
- D. $R = p + q + s + 13$

Answer to Question 22

Correct Answer: B.

This follows from taking the reciprocal of the sum of reciprocals of the resistors in a parallel circuit.

Question 21

In the given equation, b is an integer. The equation has no real solutions. What is the least possible value of b ?

$$-x^2 + bx - 169 = 0$$

- A. 26
- B. 25
- C. -26
- D. -25

Answer to Question 21

Correct Answer: D.

The discriminant $b^2 - 4ac$ must be less than 0 for no real solutions. Solve for minimum b such that $b^2 < 4(-1)(-169)$.

Question 19

The exponential function f is defined by $f(x) = ab^x$, where a and b are positive constants. If $f(n-1) = f(n) + \frac{82}{100}f(n-1)$, where n is a constant, what is the value of b ?

- A. 1
- B. 0.82
- C. 0.18
- D. 10

Answer to Question 19

Correct Answer: C.

Use the equation recursively to isolate b and simplify to find the correct constant multiplier.

Question 13

A rectangle is inscribed in a circle such that the length of the diagonal of the rectangle is twice the length of its shortest side. The circumference of the circle is 114π units. What is the area, in square units, of the rectangle?

- A. $57\sqrt{2}$
- B. $57\sqrt{3}$
- C. $3,249\sqrt{2}$
- D. $3,249\sqrt{3}$

Answer to Question 13

Correct Answer: D.

Use the relationship between diagonal and side length, then relate it to the radius (half the diagonal) from the circumference to solve for area.

Question 15

In triangle PQR , the measure of angle P is $(3x + 5)^\circ$, the measure of angle Q is $(2x + 9)^\circ$, and the measure of angle R is $(4y + 5)^\circ$. If side QR is extended through point R to point S , and the measure of angle PRS is $(x + y)^\circ$, what is the value of $x + y$?

- A. 39
- B. 29
- C. 141
- D. 146

Answer to Question 15

Correct Answer: A.

Question 11

The manager of a gym selected a sample of 139 members at random to estimate the percentage of the gym's members that would continue to pay for a membership if the price increased. From the survey, the manager estimates that 82% of the gym's members would continue to pay, with a margin of error of 6.39%. If the survey is repeated with a random sample of 278 members and the results are calculated the same way, what is the most likely effect of using the larger sample?

- A. The margin of error will be lower.
- B. The margin of error will be higher.
- C. The estimate will be lower.
- D. The estimate will be higher.

Question 18

Given the expression $\sqrt[5]{119n(\sqrt[6]{119n})^2}$, for what value of x is this equivalent to $(119n)^{30x}$, where $n > 1$?

A. 225

B. $\frac{8}{225}$

C. $\frac{4}{225}$

D. $\frac{2}{225}$

Answer to Question 18

Correct Answer: D.

Question 21

The mass of object A is 483% of the mass of object B, and the mass of object A is 0.069% of the mass of object C. If the mass of object C is $p\%$ of the mass of object B, what is the value of $\frac{p}{1,000}$?

- A. 0.7%
- B. 70
- C. 70%
- D. 700

Answer to Question 21

Correct Answer: D.

Question 22

For triangle PQR and triangle STU , angles P and S each measure 30° , with $QR = 16$ and $TU = 12$. Which additional piece(s) of information is(are) sufficient to prove triangle PQR is similar to triangle STU ?

- I. $Q = 40^\circ$, $U = 110^\circ$
 - II. $\frac{PR}{SU} = \frac{4}{3}$
- A. I is sufficient, but II is not.
- B. II is sufficient, but I is not.
- C. I is sufficient, and II is sufficient.
- D. Neither I nor II is sufficient.